



UNIVERSITEIT VAN PRETORIA  
UNIVERSITY OF PRETORIA  
YUNIBESITHI YA PRETORIA

# Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en  
Inligtingtegnologie / Lefapha la Boetšenere,  
Tikologo ya Kago le Theknolotši ya Tshedimošo

## Unit 3: Value Chain Design

BSS 310 Engineering Management 2026

Lecturer: Prof. Michael Ayomoh



# Quick Recap/Questions arising from Previous Lecture

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**The lecture addressed Market & Competitors. It focused on:**

- Understanding the principles of strategic market gap identification & market analysis in a business system and being able to deploy this on a basic level.
- Understanding the interconnectedness of the dynamic sub-entities that drives the market. These are often referred to as the market forces.

# Expected Learning Outcomes for the Current Lecture

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The expected learning outcomes herein include:

- Gaining a clear understanding of “value chain” creation in the operation of a business.
- Understanding, application, analysis, and evaluation of the SIPOC and Porter’s value chain concepts in businesses;
- Generation of the primary and support activities described in Porter’s value chain.
- Being able to link the concept of value chain to the red and blue ocean concepts via applications to diverse real-world scenarios.

# Presentation Outline

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- Define value chain
- Explore measures of introducing value chain to businesses using SIPOC and Porter's Concept
- Generation and application of the primary and support activities described in Porter's value chain
- Discuss the red and blue ocean concepts and apply these in real problems, including deployment in the semester project

# Value Chain Definition

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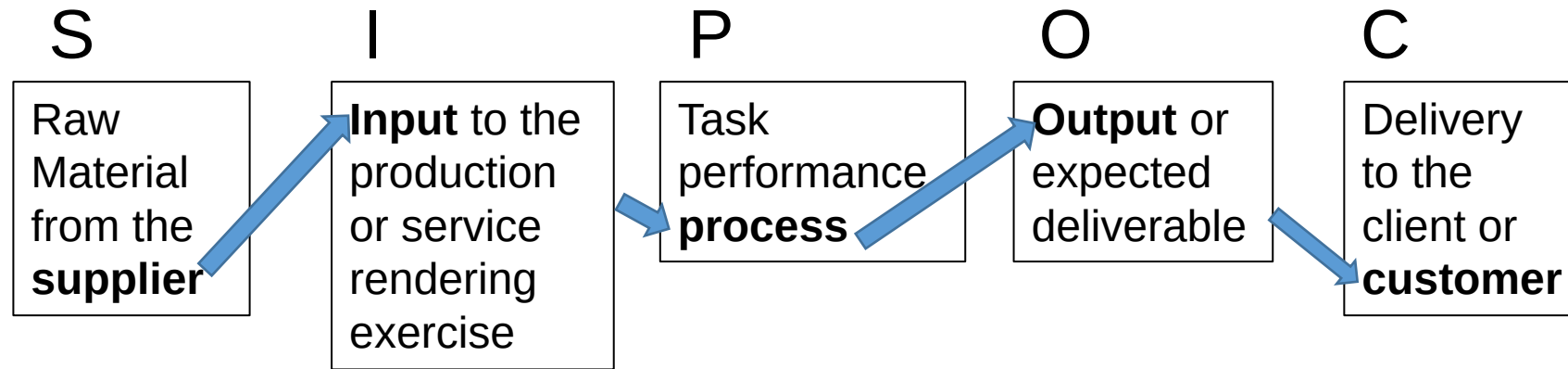
Value chain focuses on a set of activities that an organization carries out to create extra attractive gains for its customers.

A value chain is an end to end connectedness or flow of organisational specific/unique add-ons to their operations, services, products etc from the point of origination to the point of utilisation/consumption.

# The SIPOC Approach

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SIPOC- An acronym for the [(Supplier- Input-Process-Output-Customer)] interactive approach



| SIPOC                                  |                                                                |                                                                                                                                                                                                                                                                                                          |                                                                   |                                                                  |
|----------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------|
| Suppliers                              | Inputs                                                         | Process                                                                                                                                                                                                                                                                                                  | Outputs                                                           | Customers                                                        |
| (resource provider)                    | (process)                                                      | (high level process flow)                                                                                                                                                                                                                                                                                | (from the process)                                                | (receiver an output from the process)                            |
| Coffeemaker purchased - on countertop  | >5 cup capacity coffee maker                                   | <b>Making coffee</b><br><br><div> <div>Add water</div> <div>↓</div> <div>Add filter &amp; ground coffee</div> <div>↓</div> <div>Plug-in and turn on</div> <div>↓</div> <div>Pour into mug</div> <div>↓</div> <div>Add condiments</div> <div>↓</div> <div>Stir</div> <div>↓</div> <div>Serve</div> </div> | heating to keep coffee warm for 1 hour after brewing              | All of us enjoy the same brand of coffee with varying condiments |
| city water supply into faucet          | water supply                                                   |                                                                                                                                                                                                                                                                                                          | enough coffee to serve all of us within 15 minutes of start time. | 1 cup of coffee                                                  |
| purchase from XYZ company              | 1 filter                                                       |                                                                                                                                                                                                                                                                                                          | one filter to prevent overflow                                    | 1 teaspoon of french vanilla creamer                             |
| Use ABC brand beans                    | 4 tablespoons of coffee grinds                                 |                                                                                                                                                                                                                                                                                                          | correct amount of grinds                                          |                                                                  |
| Electric company                       | 120V GFCI outlet                                               |                                                                                                                                                                                                                                                                                                          |                                                                   | 1 cup of coffee                                                  |
| Upper left drawer next to refrigerator | measuring spoons                                               |                                                                                                                                                                                                                                                                                                          | Source to heat water to temperature                               | Honey on the table                                               |
|                                        |                                                                |                                                                                                                                                                                                                                                                                                          | Pump to move water up through filter.                             | Dash of cinnamon                                                 |
| Mugs purchased - in upper left cabinet | coffee mugs                                                    |                                                                                                                                                                                                                                                                                                          | Hot coffee filled near the top of the mug.                        |                                                                  |
|                                        |                                                                |                                                                                                                                                                                                                                                                                                          |                                                                   |                                                                  |
| refrigerator and pantry                | condiments and containers for sugar, creamer, honey, cinnamon. |                                                                                                                                                                                                                                                                                                          |                                                                   | 1 small cup of coffee for each                                   |
| ABC brand for sugar & creamer.         |                                                                |                                                                                                                                                                                                                                                                                                          | coffee served in spouses favorite mug                             | 1 teaspoon sugar                                                 |
| XYZ brand for honey and cinnamon       |                                                                |                                                                                                                                                                                                                                                                                                          | coffee served in husbands mug                                     | 1 tablespoon french vanilla creamer                              |
| pantry                                 | stirrers, lids                                                 |                                                                                                                                                                                                                                                                                                          | coffee served in personalized kids mugs                           | Let sit for 5 minutes before serving                             |

2 teenagers

# Porter's Value Chain

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The Porter's Value Chain is a useful strategic management tool. It works by breaking an organization's activities down into strategic pieces, for holistic visualisation of the entire picture of the cost drivers, sources of differentiation and then make changes appropriately.



# Porter's Value Chain: Understanding How Value Is Created in Businesses

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- **How does your organization create value?**
- **How do you change business inputs into business outputs** in such a way that they have a greater value than the original cost of creating those outputs?
- This isn't just a dry question: it's a matter of fundamental importance to companies, because it addresses the economic logic of why the organization exists in the first place.

# **How Value Is Created in Businesses: Porter's approach**

# Porter's Value Chain: Understanding How Value is Created in Businesses

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- Manufacturing companies
- Retail businesses
- Education
- Restaurant Services

# Porter's Value Chain: Understanding How Value Is Created in Businesses

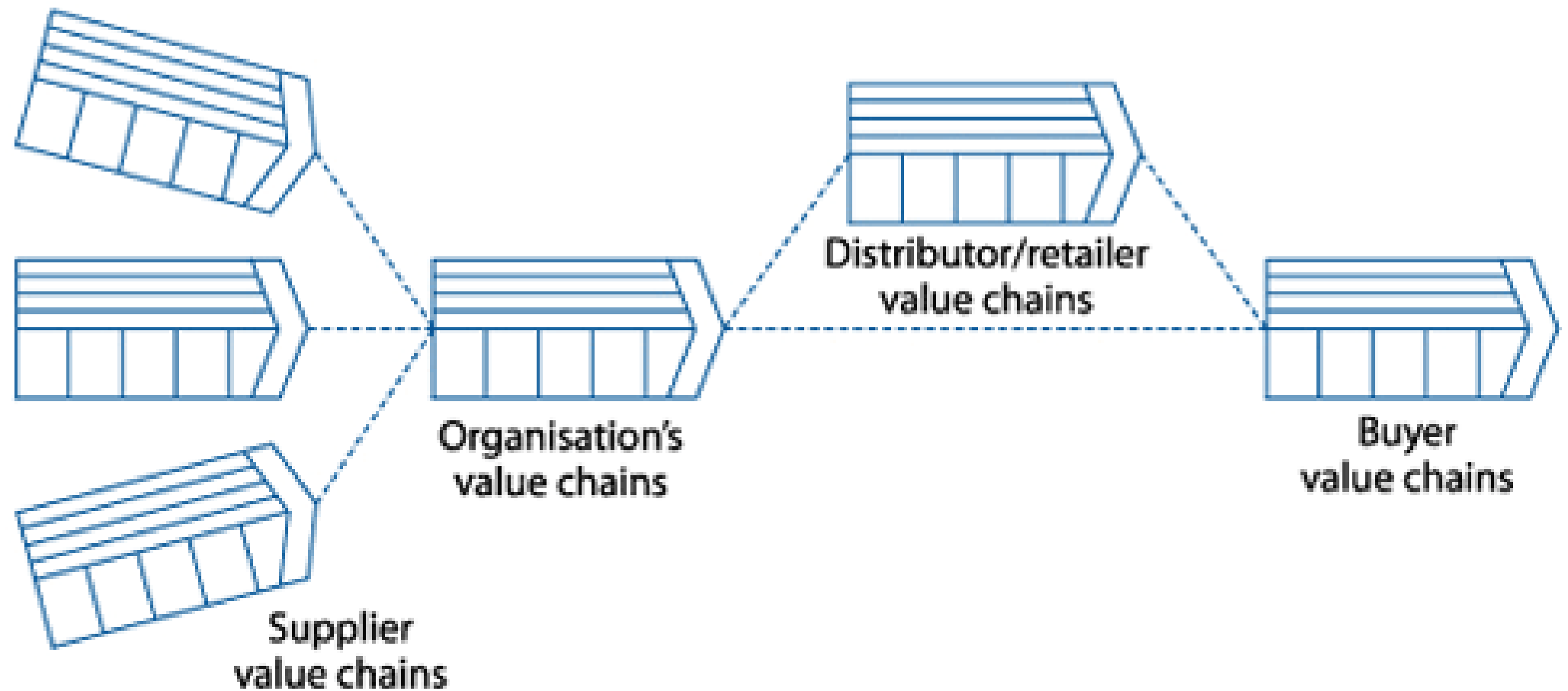
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**Value Created and Captured – Cost of Creating that Value = Margin**

Understanding how your company creates value, and looking for ways to add more value, are critical elements in developing a competitive strategy. Michael Porter discussed this in his influential 1985 book "[Competitive Advantage](#),"



Element in Porter's Value Chain



## Value Chain Network

# Using Porter's Value Chain to identify and understand a company's value chain

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## Step 1 – Identify sub-activities for each primary activity[Book publishing service]

- **Direct activities** create value by themselves. For example, in a book publisher's marketing and sales activity, direct sub-activities include: typesetting and formatting, infographing etc.
- **Indirect activities** allow direct activities to run smoothly. For the book publisher's sales and marketing activity, indirect sub-activities include: sales calls to bookstores, advertising, selling online, managing the sales force and keeping customer records etc.

# Using Porter's Value Chain to identify and understand a company's value chain

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- **Quality assurance:** This activity ensure that direct and indirect activities meet the necessary standards. For the book publisher's sales and marketing activity, this might include proof-reading and editing amongst others.



# Using Porter's Value Chain to identify and understand a company's value chain

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**Step 1 – Identify sub-activities for each primary activity**  
**[Product Manufacturing]** **Direct activities** create value by themselves. For example, in a manufacturing business, direct sub-activities include processing of raw materials (drilling, grinding, boring, polishing, heating etc), packaging, provision of a stable power supply etc.

- **Indirect activities** allow direct activities to run smoothly. For the manufacturing business, this can include maintenance of the machines, shop floor inventory monitoring, package labelling etc.

# Using Porter's Value Chain to identify and understand a company's value chain

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- **Quality assurance:** This activity ensure that direct and indirect activities meet the necessary standards. For the manufacturing operation, this might include product test for reliability, physical defect tracking and prevention etc.

# Using Porter's Value Chain to identify and understand a company's value chain

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**Step 2 – Identify sub-activities for each support activity**  
spread across: direct, indirect and quality checks e.g. admin, security, logistics etc.

# Using Porter's Value Chain to identify and understand a company's value chain

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## Step 3 – Identify links

Find the connections between all the value activities you've identified.

# Using Porter's Value Chain to identify and understand a company's value chain

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## Step 4 – Look for opportunities to increase value

- **Review each of the sub-activities and links** whether you are trying to set yourself apart from your competitors or simply having a lower cost base.
- **a need for prioritisation** resulting from a huge list of changes.

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# **Value Chain: The Red Ocean vs the Blue Ocean scenarios**

# Value Chain: The Red Ocean vs the Blue Ocean scenarios

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Value chain can be a means for a Blue Ocean business sustainability.

Value chain can aid in converting Red Ocean businesses/project scenarios into a Blue Ocean Success.

# Value Chain: Red Ocean vs Blue Ocean

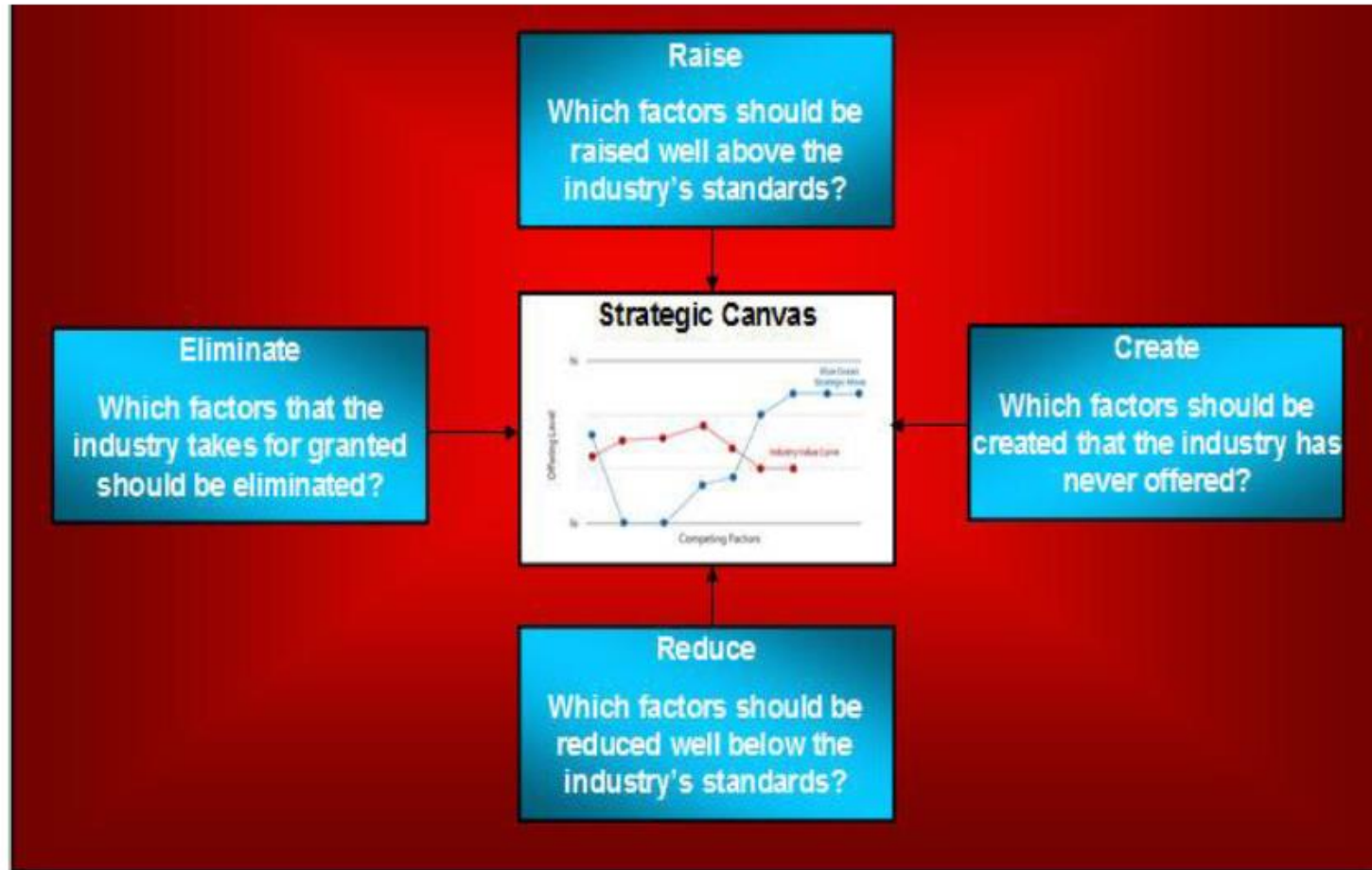
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| <b>Red Ocean Strategy</b><br>Focus on current customers                                                         | <b>Blue Ocean Strategy</b><br>Focus on noncustomers                                               |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| • Compete in existing markets                                                                                   | • Create uncontested markets to serve                                                             |
| • Beat the competition                                                                                          | • Make the competition irrelevant                                                                 |
| • Exploit existing demand                                                                                       | • Create and capture new demand                                                                   |
| • Make the value-cost trade-off                                                                                 | • Break the value-cost trade-off                                                                  |
| • Align the whole system of a firm's activities with its strategic choice of differentiation <u>OR</u> low cost | • Align the whole system of a firm's activities in pursuit of differentiation <u>AND</u> low cost |



# Value Chain: Some Actions and Strategies to transition from a Red to a Blue Ocean Business

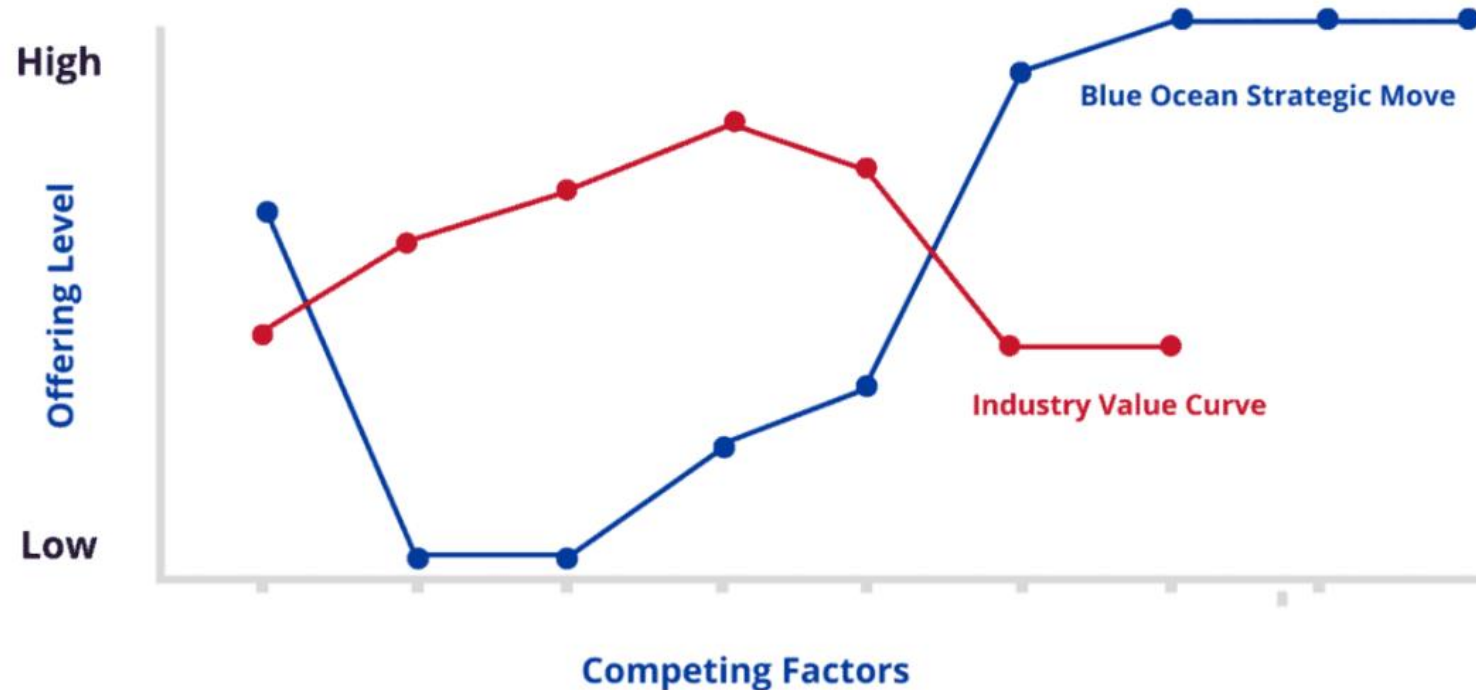
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- <https://www.blueoceanstrategy.com/tools/strategy-canvas/>

# Value Chain: Some Actions and Strategies to transition from a Red to a Blue Ocean Business

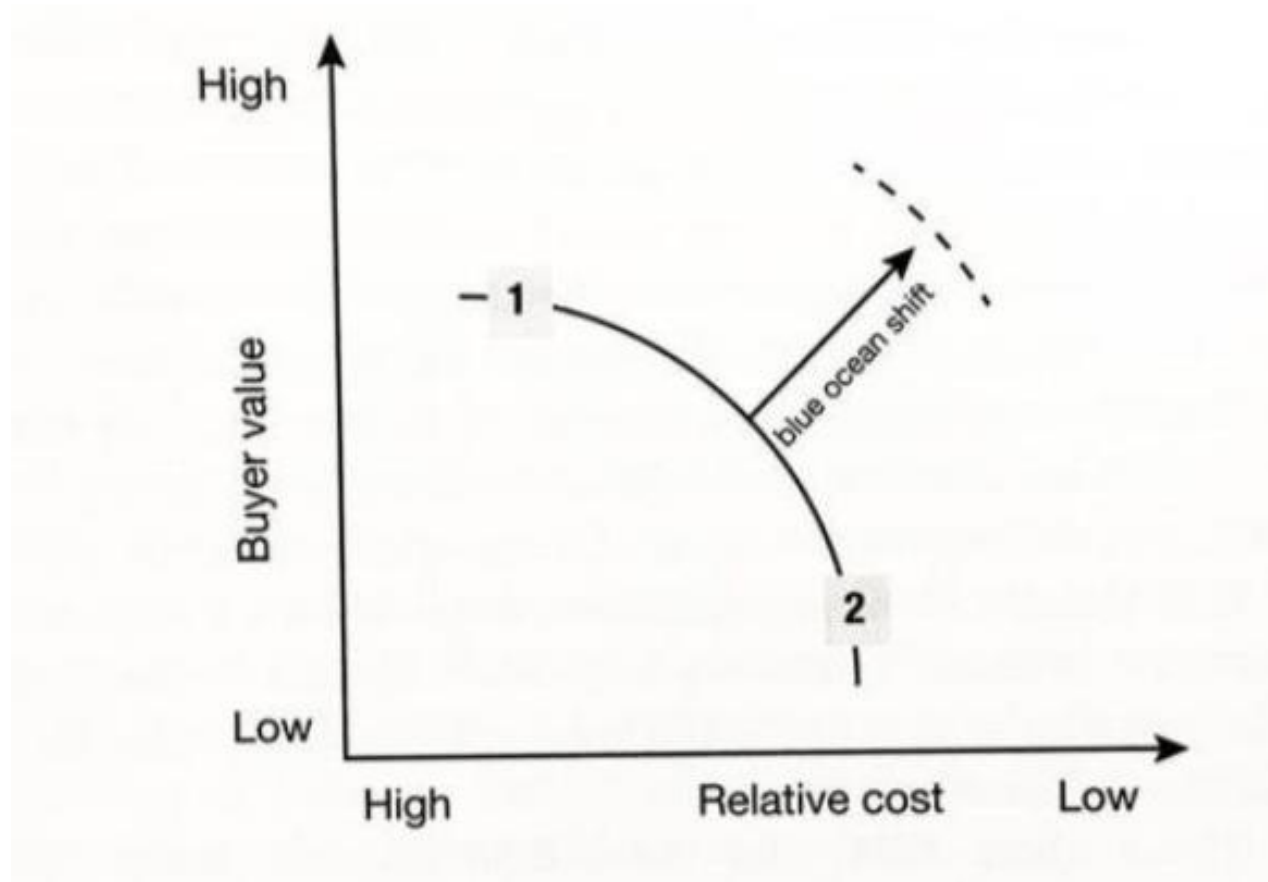
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- <https://www.blueoceanstrategy.com/tools/strategy-canvas/>

# Value Chain: A Blue Ocean Shift

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# Source

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S. Rezeki, I. G. E. P. S. Sentanu, B. Sanawiri, K. Shankar, P. T. Nguyen, (2019), **Blue Ocean Strategy for creating Value of the Organization: Examination of Differentiation with Red Ocean Strategy**, *International Journal of Recent Technology and Engineering* (IJRTE) ISSN: 2277-3878, Volume-8, Issue-2S11, September 2019

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# Source

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<https://www.investopedia.com/articles/investing/103114/starbucks-example-value-chain-model.asp>

<https://www.six-sigma-material.com/>

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**End**



# Next lecture

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## Unit 4: Production Systems



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